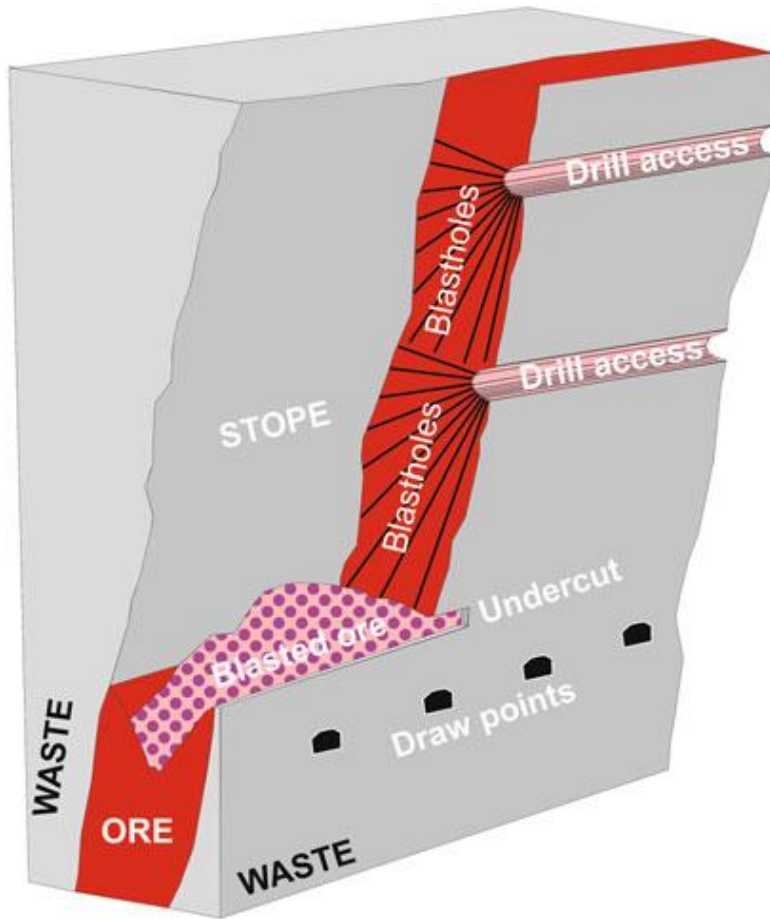


Sublevel stoping with backfilling

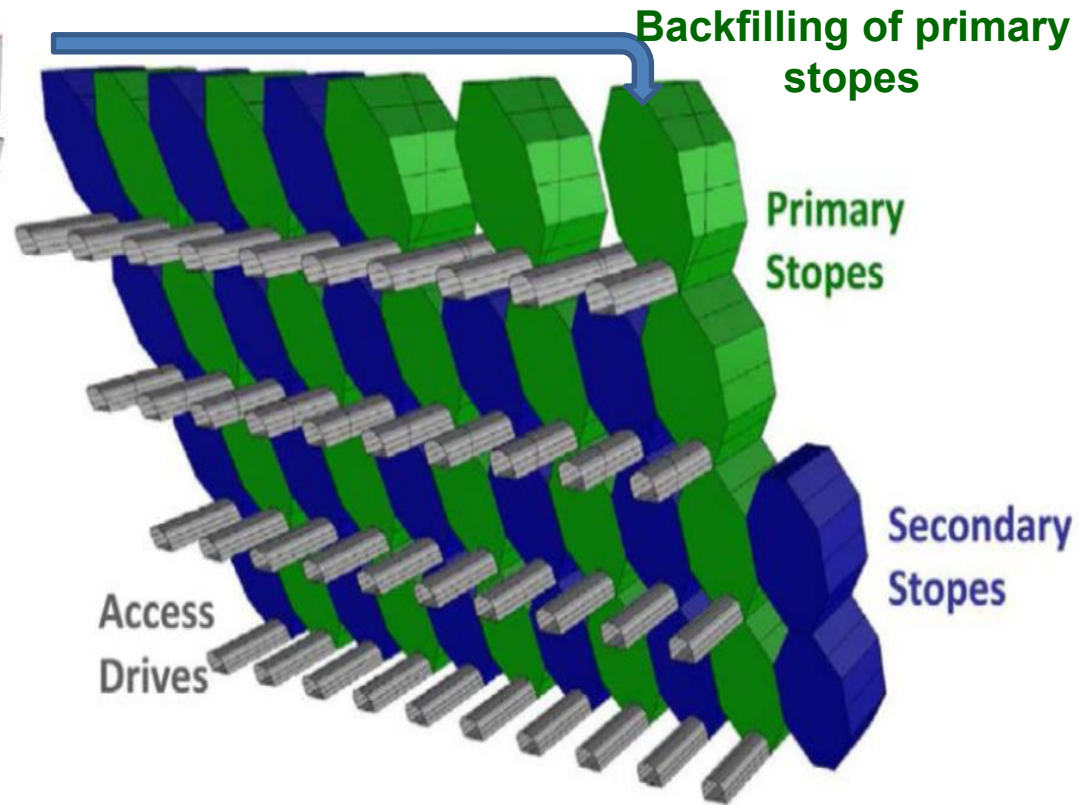
SUITABLE CONDITIONS

- Strong ore and wall rocks
- Dip: steep ($>60^\circ$) (greater than angle of repose-gravity flow)
- Smooth ore to wall rock contact: ease in movement of material.
- Competent ore and host rock.
- Regular ore bodies and method is non-selective

(a) Sublevel stoping



(b) Primary-secondary stope extraction



- Drilling shorter blast holes at regular vertical intervals
- Ore transportation is always done from the bottom most draw level

Production drilling and blasting

- Larger diameter drilling and blasting
- DTH
- Hole depth: Depends on stope height
- Drilling in advance of blasting face.

Explosives

- Cartridge emulsion/Slurry
- Site mixed emulsion (SME)/site mixed slurry (SMS)
- Charging down hole or up hole or combination of both
- Compressed ANFO used for up hole charging, compression provides adhesiveness to explosive to remain within blast hole.
- High viscous SME/SMS for up hole charging
- Ease in pouring downhole
- Ease in pumping up hole.
- Ring blasting pattern

Loading and Hauling

1. LHD (Remotely operated)
2. LHD loading from draw points
3. Ore chutes-mine cars/tubs
4. Transportation: LPDT



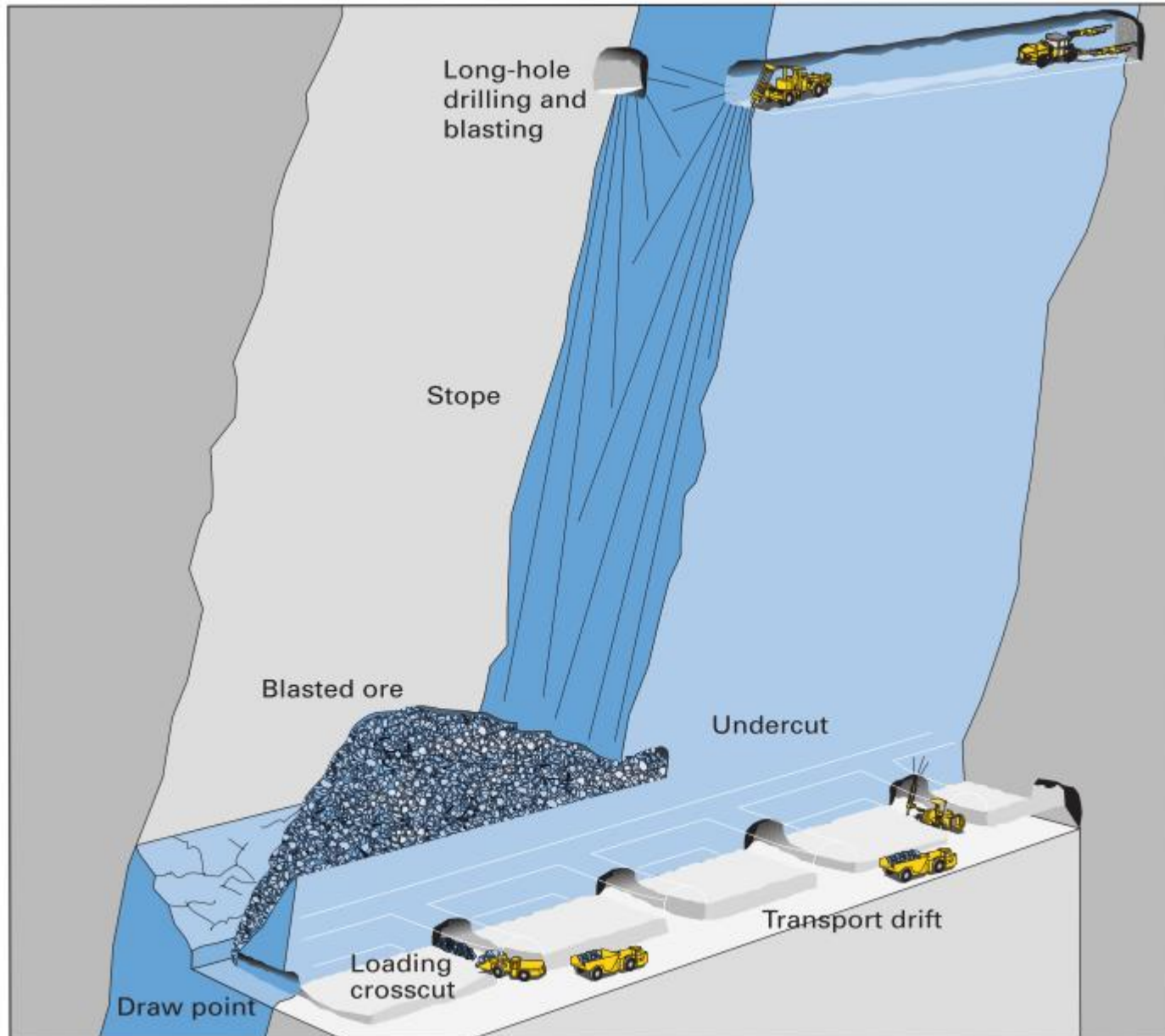
Ground control

1. Not a problem –strong ore body
2. Cable bolting (6m) during drilling long holes.(resin or cement grouted)
3. Rock bolt (2.4m)
4. Wire mesh
5. Shotcrete
6. Backfilling (Paste/Hydro fill/CRF/Waste rock fill in secondary)

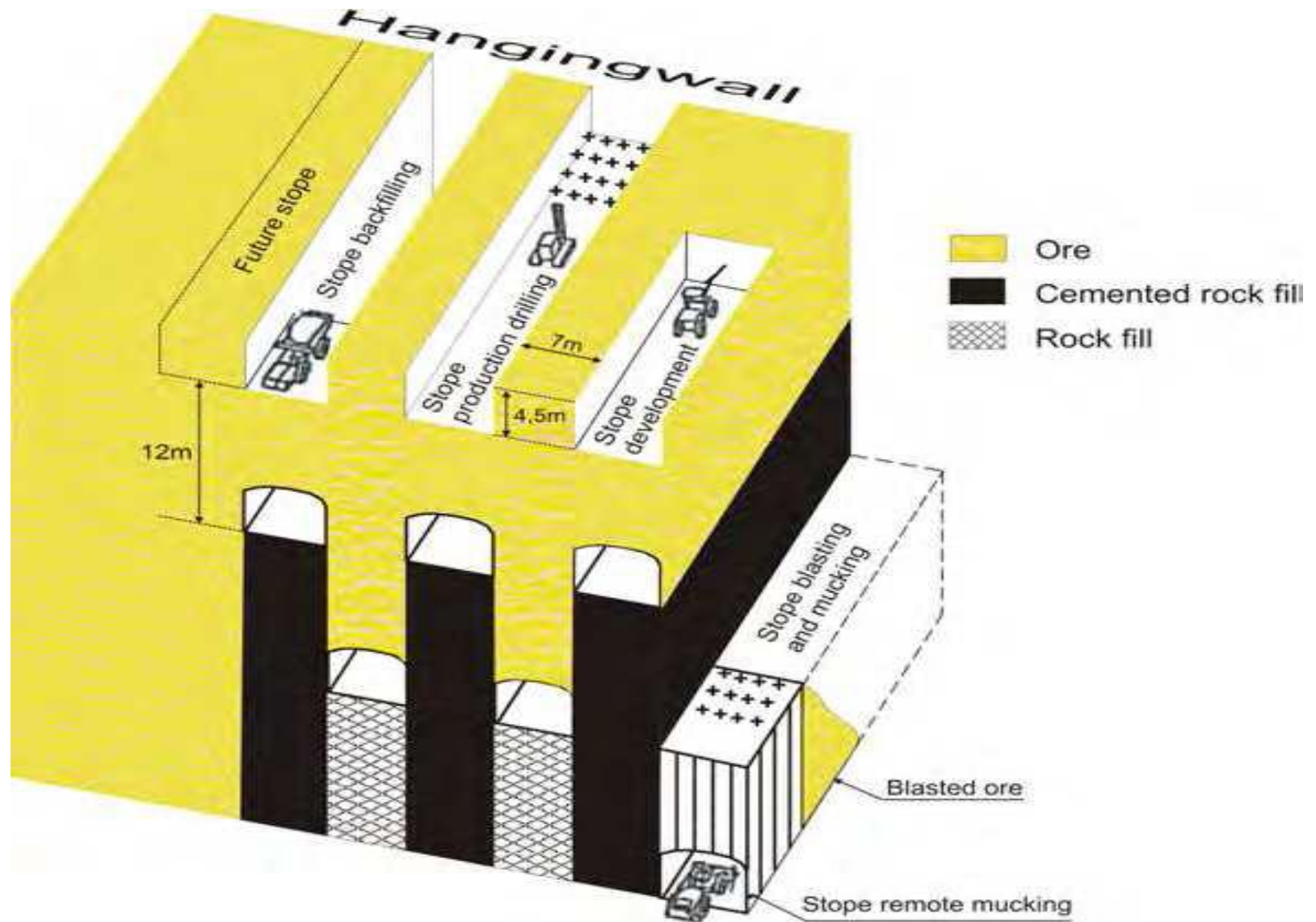
Long hole stoping with back filling

- In massive, regular, and large ore bodies, selective mining may not be appropriate from the cost perspective.
- Large size underground mining machineries and bulk mining method like long hole/blast hole stoping may be used under such circumstances.
- Hence, a large volume of ore can be extracted with high productivity.
- Long hole stoping, a variant of sublevel open stoping, is a highly productive and low-cost metalliferous mining method in which large diameter (140 to 165 mm) blast holes are used.
- The depth of the blast holes may reach up to 100 m.
- Involves drilling a series of vertical or inclined **long holes** from the **top to the bottom of a stope**.
- In this method, the miners create a vertical slot at one end of the stope and then work in the levels to drill a radial pattern of drill holes.
- After charging a set of these blast holes with explosives, the blocks of ore body are blasted. After extraction of the stopes, the primary voids are backfilled with cemented backfill, and the secondary voids are backfilled with waste rock (uncemented)

Long hole stoping



Long hole stoping with back filling



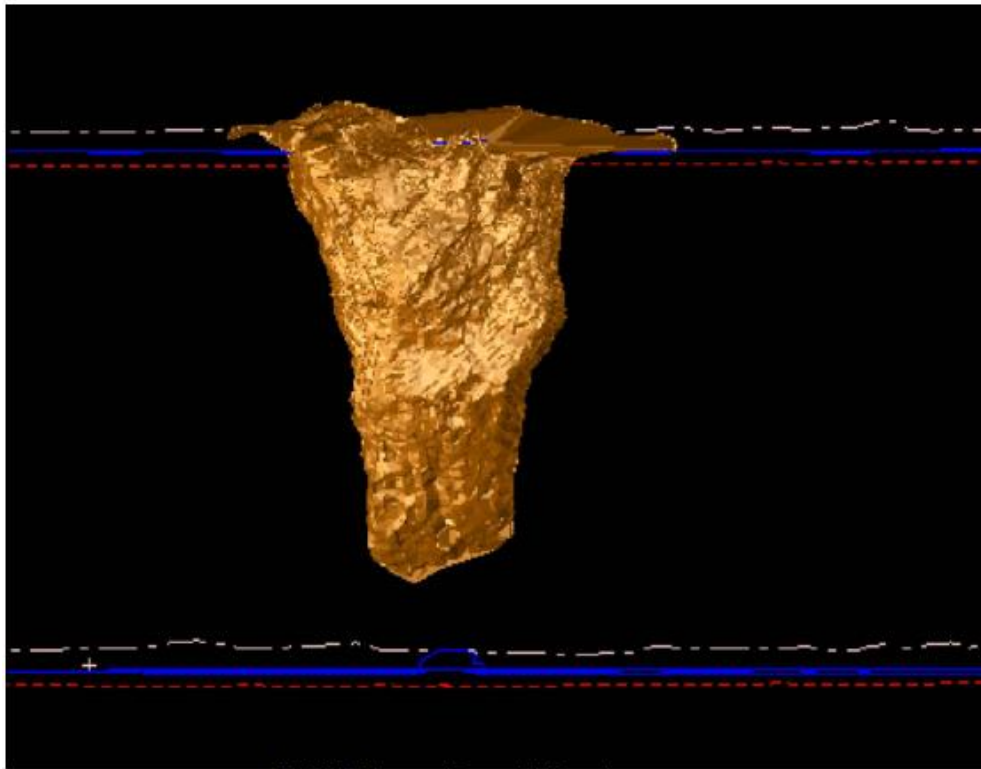
Blast hole stoping with back filling

- At drill level, the orebody is accessed from the foot wall waste drive by crosscuts.
- Within the ore body, one/two drive(s) are developed.
- At the extraction level, cross cuts are developed from the foot wall extraction drive. An ore drive is developed along the strike of the center of the ore-body as Trough Drive.
- From where upwards holes are drilled for formation of 'Trough'.
- The cross cuts between the foot wall waste drive and the trough drive are used as 'Draw Points'.
- Initially, a slot raise is developed and is further widened to make a slot between H/W and F/W.
- Downward ring drilling are done from drill level, whereas, upward trough drilling are done from trough drive.
- The rings are blasted against the slot and blasted ore is extracted through cross cuts at draw level.

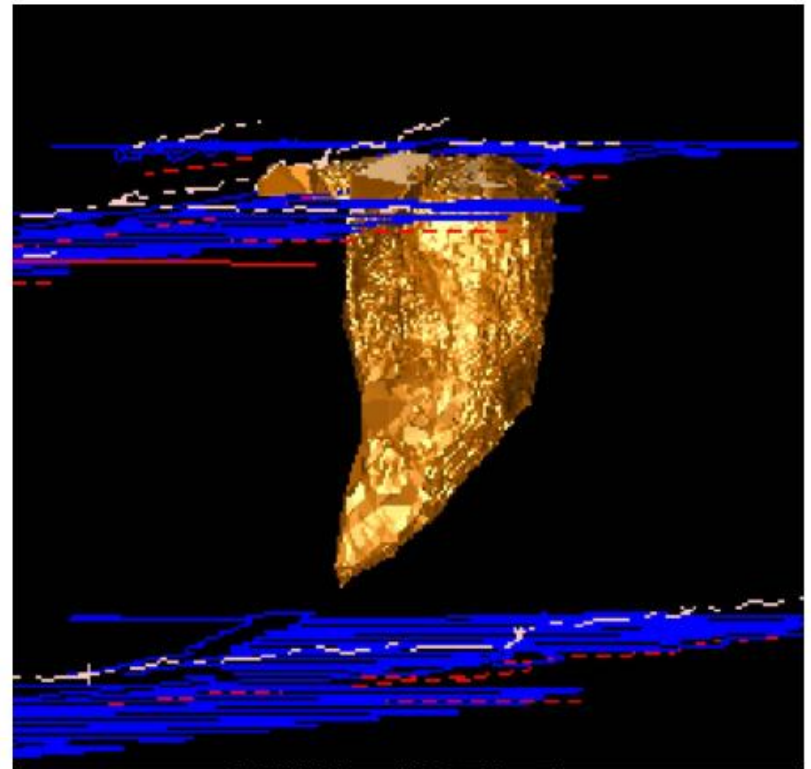
Example:

Blast hole stoping with backfilling: CMS model

AS13 Stope(72mRL to 25mRL)



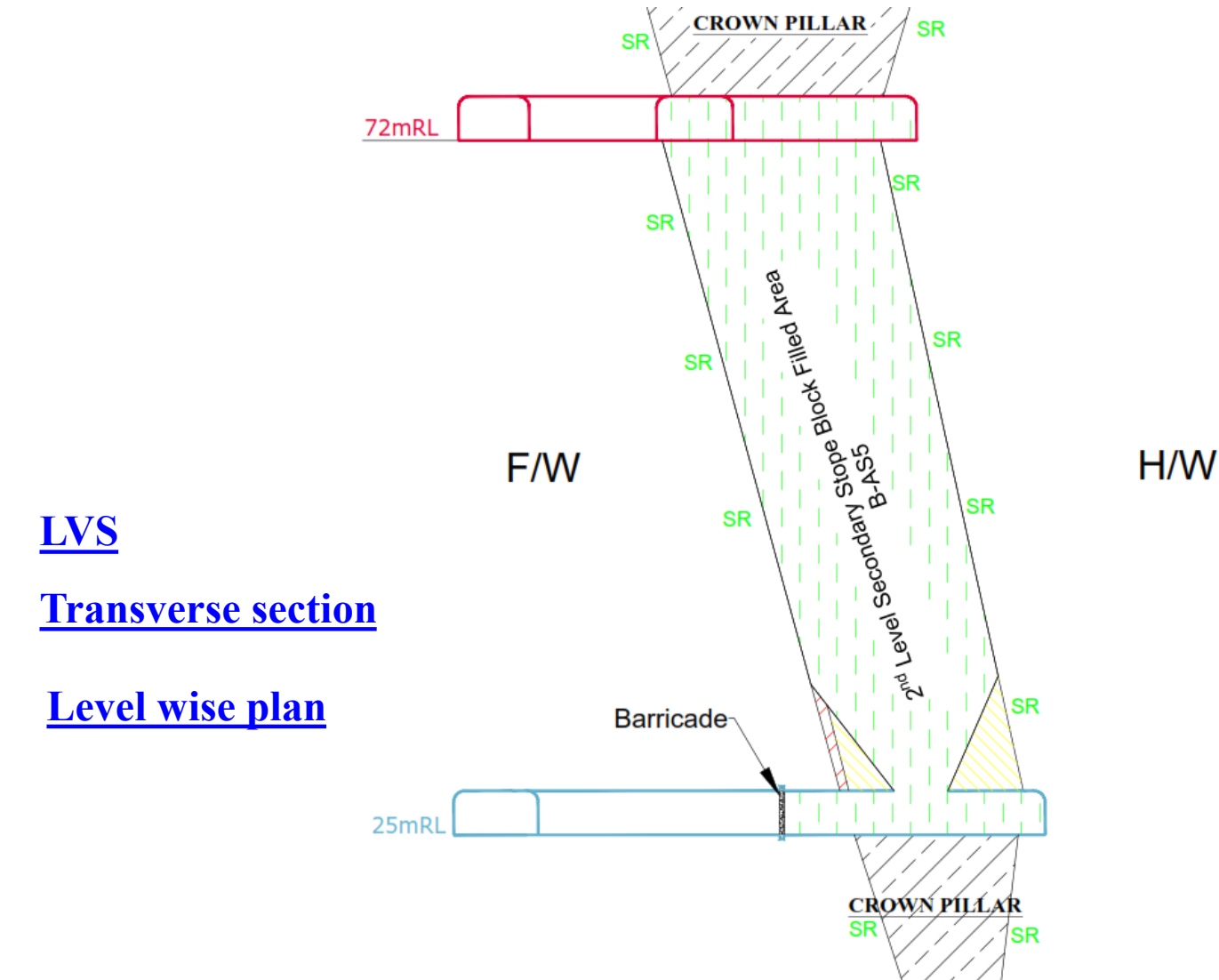
AS13 Stope(Front View)



AS13 Stope(Side View)

Example:

Blast hole stoping with backfilling

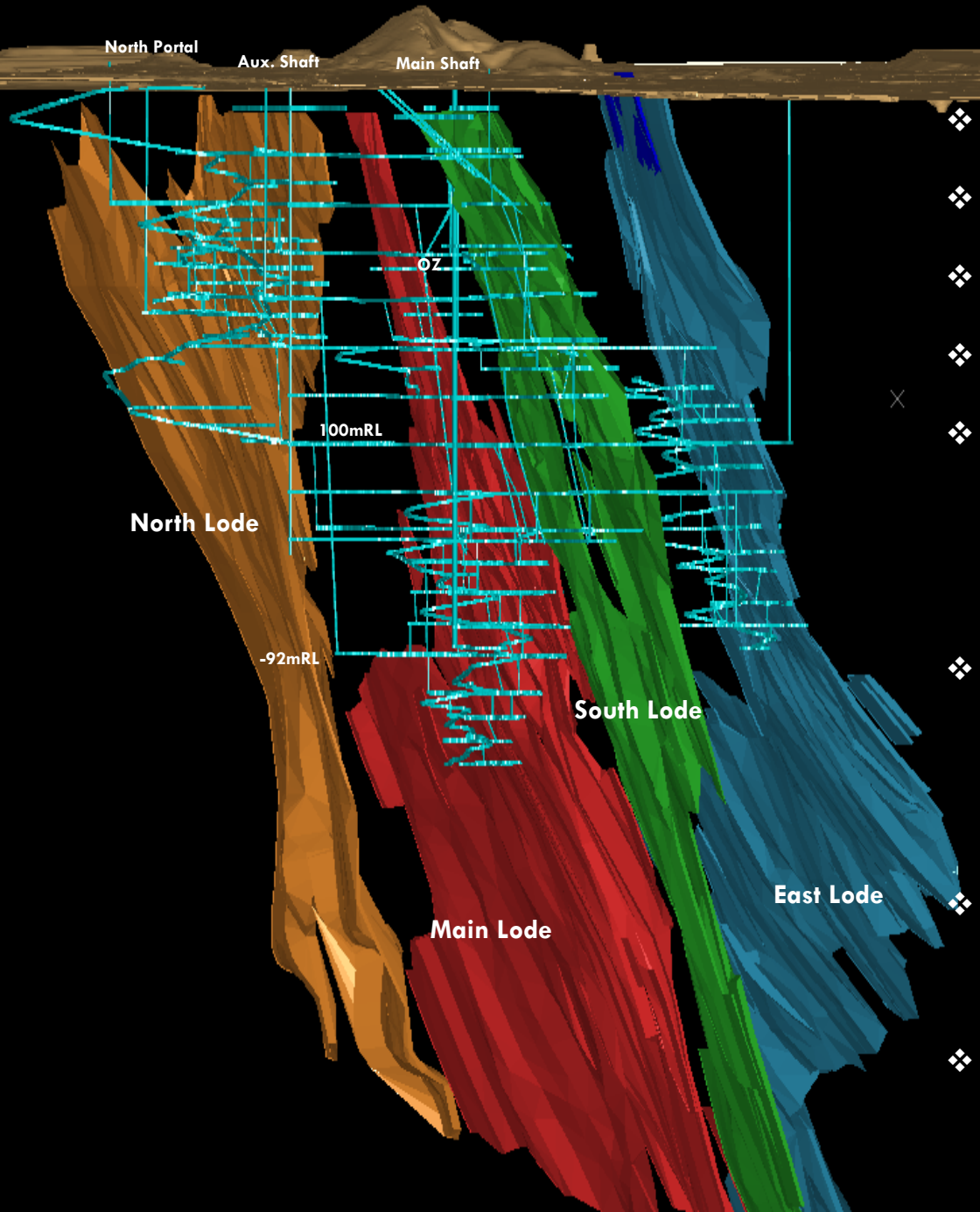


LVS

Transverse section

Level wise plan

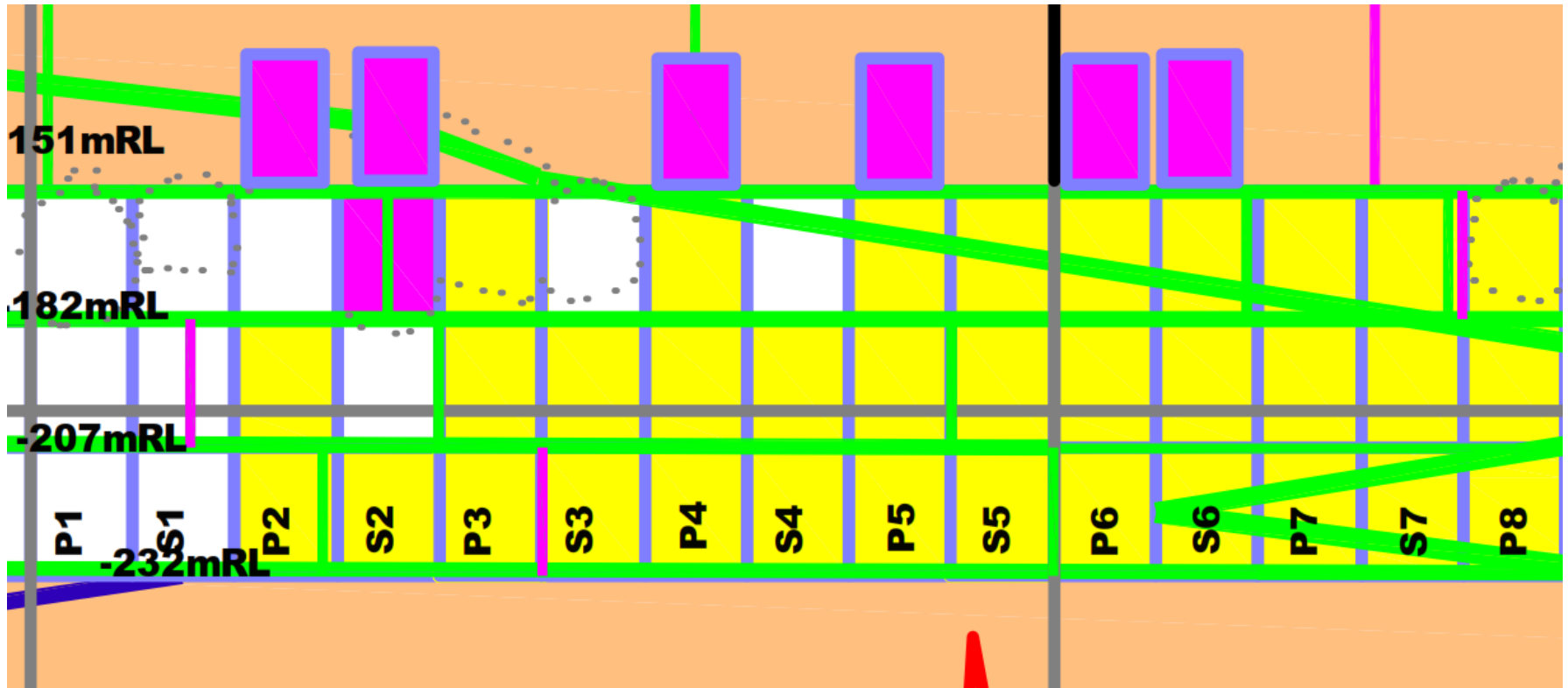
EXAMPLE



- ❖ Production since: **1983**
- ❖ Mode of Access: **2 Shafts & 1 Ramp**
- ❖ Capacity : **2.0 Mtpa**
- ❖ Ventilation: **Peripheral Type**
- ❖ Major Equipment: **30t LPDTs, 17t LHDs, Jumbo, Tower Mounted Koepe Winder (6.8t twin skips), Ground Mounted Drum Winder**
- ❖ Infrastructure: **Ug Crusher, Ug Workshops, Ventilation Raises & Fans, Sumps, Substations etc.**
- ❖ Beneficiation Plant of **1.1 Mtpa Capacity, Hydraulic Backfilling**
- ❖ Paste Fill Plant **550 TPH capacity**

Example

Blast hole stoping with backfilling



LVS

Transverse and Longitudinal stoping

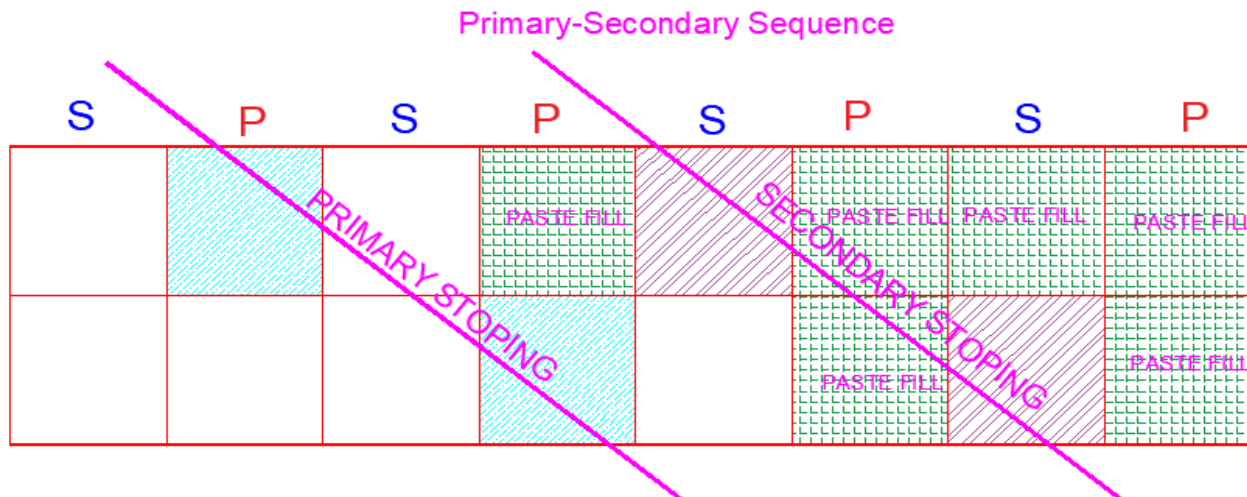
- If the ore body is mined by retreating from the hang wall (H/W) to the foot wall (F/W) for thick ore bodies.
- It is referred to as transverse stoping.
- Here, the long axis of the stopes and drifts are perpendicular to the strike of the ore body.
- In contrast, extraction from stopes along the strike of the ore body is referred to as longitudinal stoping.
- This approach is used if the thickness of the ore body is lesser and is not suitable for transverse stoping.

Underhand mining

Primary-secondary Sequence of stoping

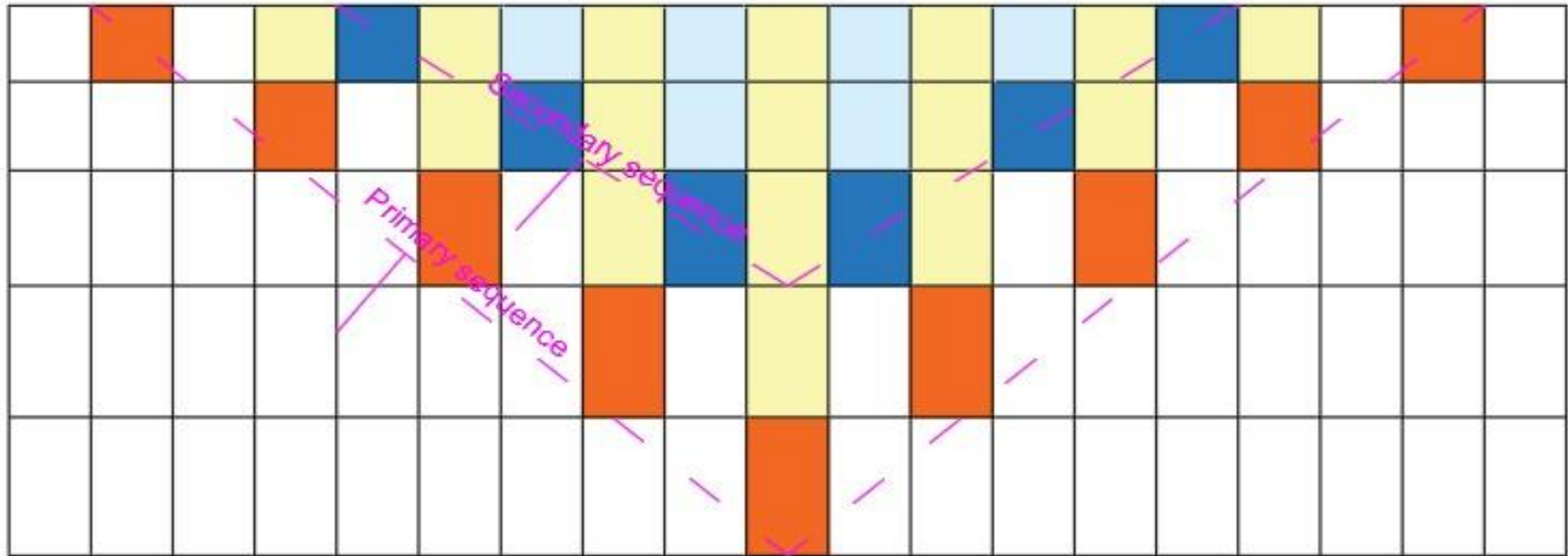
Double diagonal line of extraction with primary stoping at a lead of 2 stopes and bottom-level stoping at a lag of 1 or 2 stopes

In primary-secondary sequence, a double diagonal line of extraction is maintained, with the primary stoping at a lead of 2 stopes with respect to the secondary stoping in the same sub-level. The bottom level will have the similar primary-secondary sequence with a lag of 1 or 2 stopes with respect to the top level as shown. Thus, a double diagonal line of extraction is maintained.

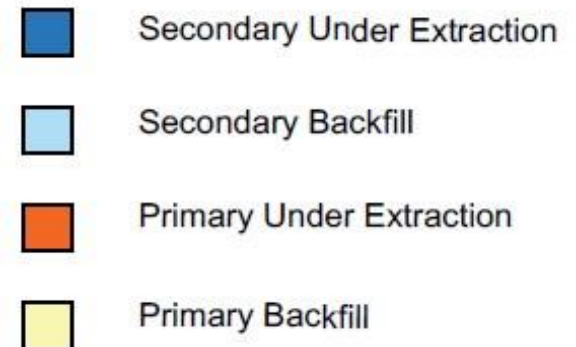


Underhand mining

Primary-secondary Sequence of stoping



Underhand sequence of mining with paste filling required re-development of roadway in paste backfill.

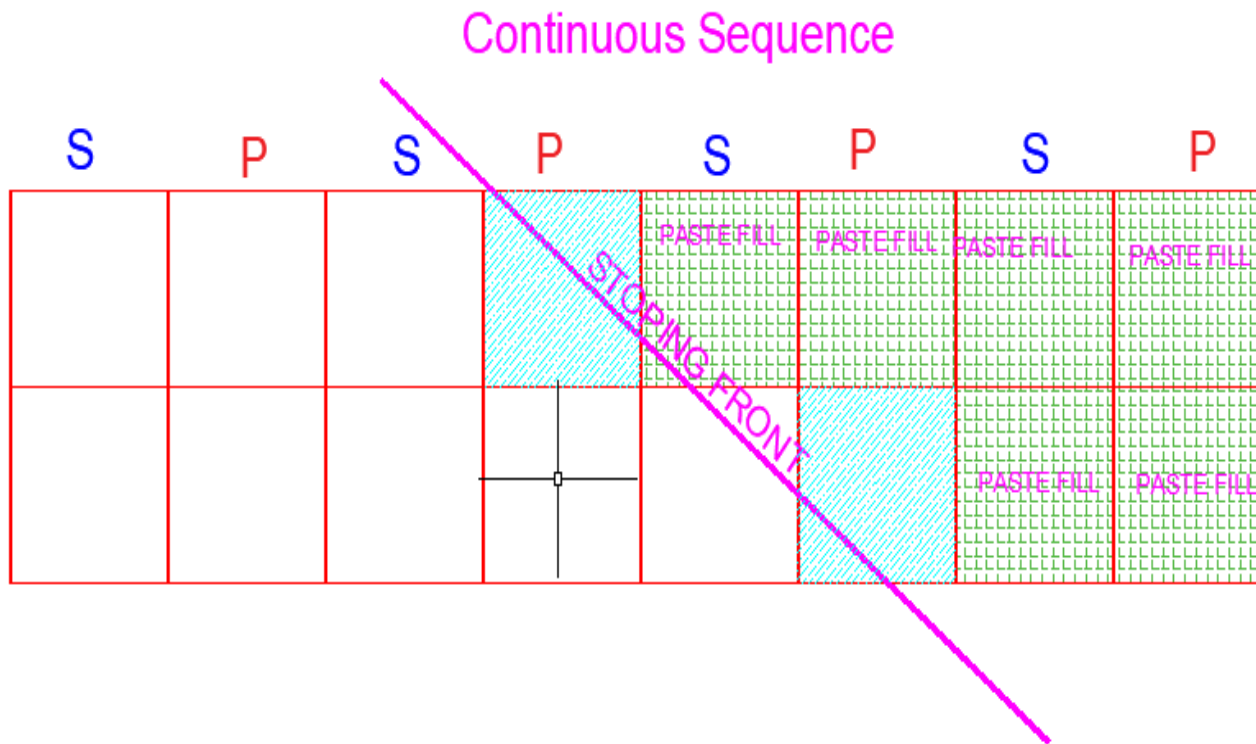


Underhand mining

Continuous sequence of stoping

Single diagonal line of extraction and bottom-level at a lag of 2 or 3 stopes

In continuous sequence, the stopes are extracted continuously, by keeping the bottom level at a lag of 2 or 3 stopes with respect to the top level

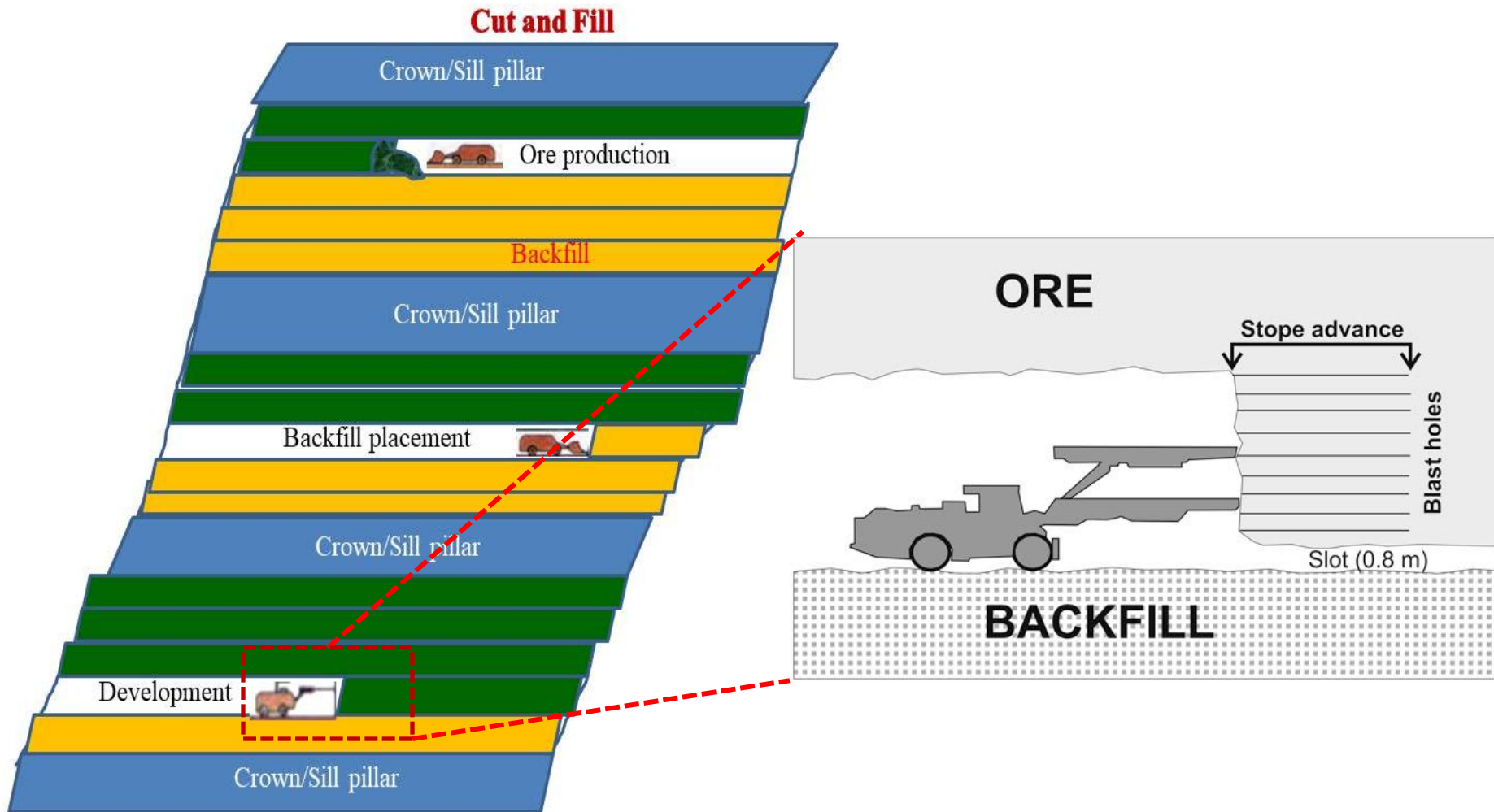


Cut and fill method

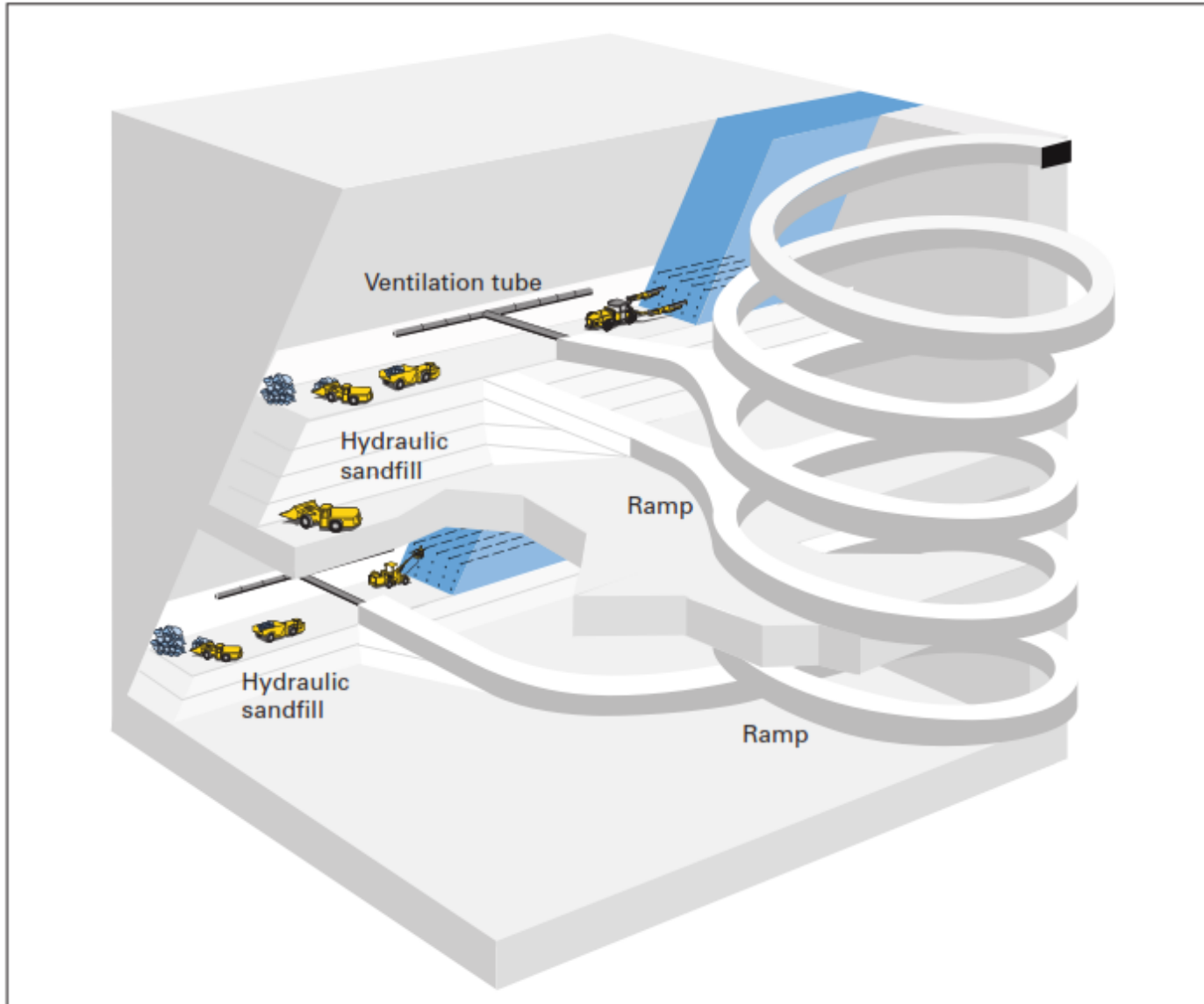
Applicability: Veins & Inclined tabular ore bodies with dips in the range of 35 - 90 degree.

- Cut and fill stoping is generally adopted in veins and inclined tabular ore bodies.
- Where the host rock mass is of insufficient strength to sustain the significant unsupported spans.
- Brady and Brown (1993) suggested that this method is suitable for ore bodies with dips in the range of 35-90°.
- In cut and fill stoping, mining advances in the bottom-up sequence, whereby an initial ore drive is developed at the bottom of a stope block.
- Backfill is then placed on the floor of the drive to act as a platform for developing another ore drive
- In contrast, the top-down mining sequence requires horizontal crown pillars to act as a barrier against the inrush of uncemented backfill from the overlying mined-out stopes.
- They also act as regional pillars for the long-term stability of the mine structures.

Cut and fill method



Cut and fill layout



Mining Method Selection

To select a mining method, certain data describing the orebody is required.

- Geological cross sections and a longitudinal section
 - Level maps
 - Block model (grade model)
 - Geomechanical characteristics of the host and surrounding rock.
1. One approach is to find one or more comparable ore bodies that are being or were mined successfully and use that mining method(s) to determine the most likely methods to investigate further.
 2. Another approach is to determine applicable mining methods and develop a short list for detailed consideration through a process of rationalization.

Following are typical considerations to be weighed in selecting a mining method (listed roughly in order of importance)

- Maximize **safety** (integrity of the mine workings as a whole or in part).
- Minimize **cost** (bulk mining methods have lower operating costs than selective extraction).
- Minimize the **schedule** required to achieve full production (optimize stope sequencing).

Cont.....

Mining Method Selection

- Optimize **recovery** (80% or greater recovery of geological reserves).
- Minimize **dilution** (20% or less dilution of waste rock that may or may not contain economic minerals).
- Minimize **stope turn around** (cycle) time (drill, load, blast, muck, backfill, set).
- Maximize **mechanization** (trackless versus track).
- Maximize **automation** (employment of remote controlled LHDs).
- Minimize **pre-production development** (top down versus bottom up mining).
- Minimize **stope development** (selective versus bulk mining methods).
- Maximize **gravity assist** (underhand versus overhand).
- Maximize **natural support** (partial extraction versus complete extraction).
- Minimize **retention time** of broken ore (open stoping versus shrinkage).
- Maximize **flexibility and adaptability** based on size, shape, and distribution of target mining areas.
- Maximize **flexibility and adaptability** based on distribution and variability of ore grades.
- Maximize **flexibility and adaptability** to sustain the mining rate for the mine life.
- Maximize **flexibility and adaptability** based on access requirements.
- Maximize **flexibility and adaptability** based on opening stability, ground support requirements, hydrology (ground water and surface runoff), and surface subsidence.

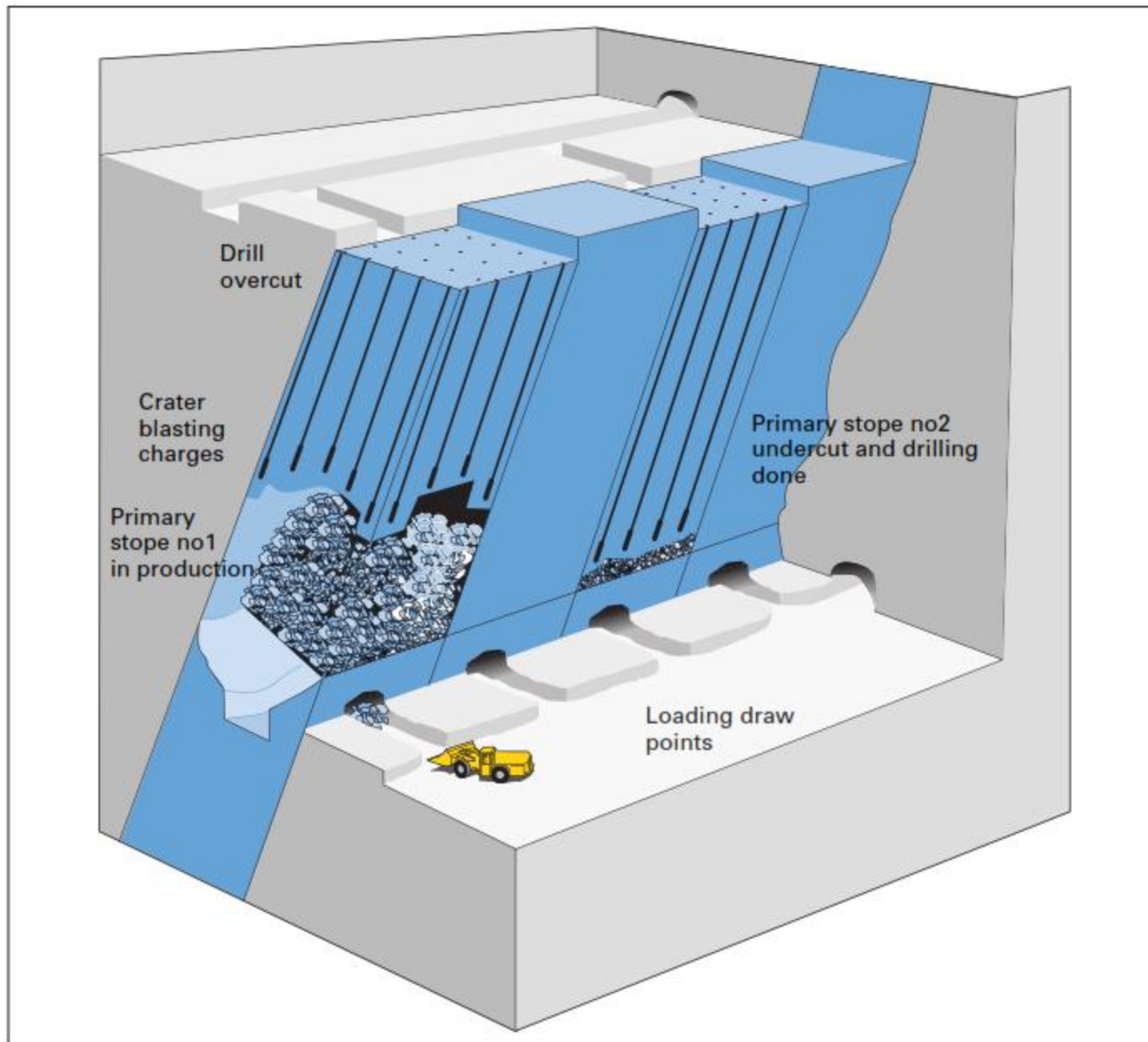
Underground infrastructure: Shaft, Decline, Drifts, Ore drives, Ore pass, ore bin, Levels, Sub levels

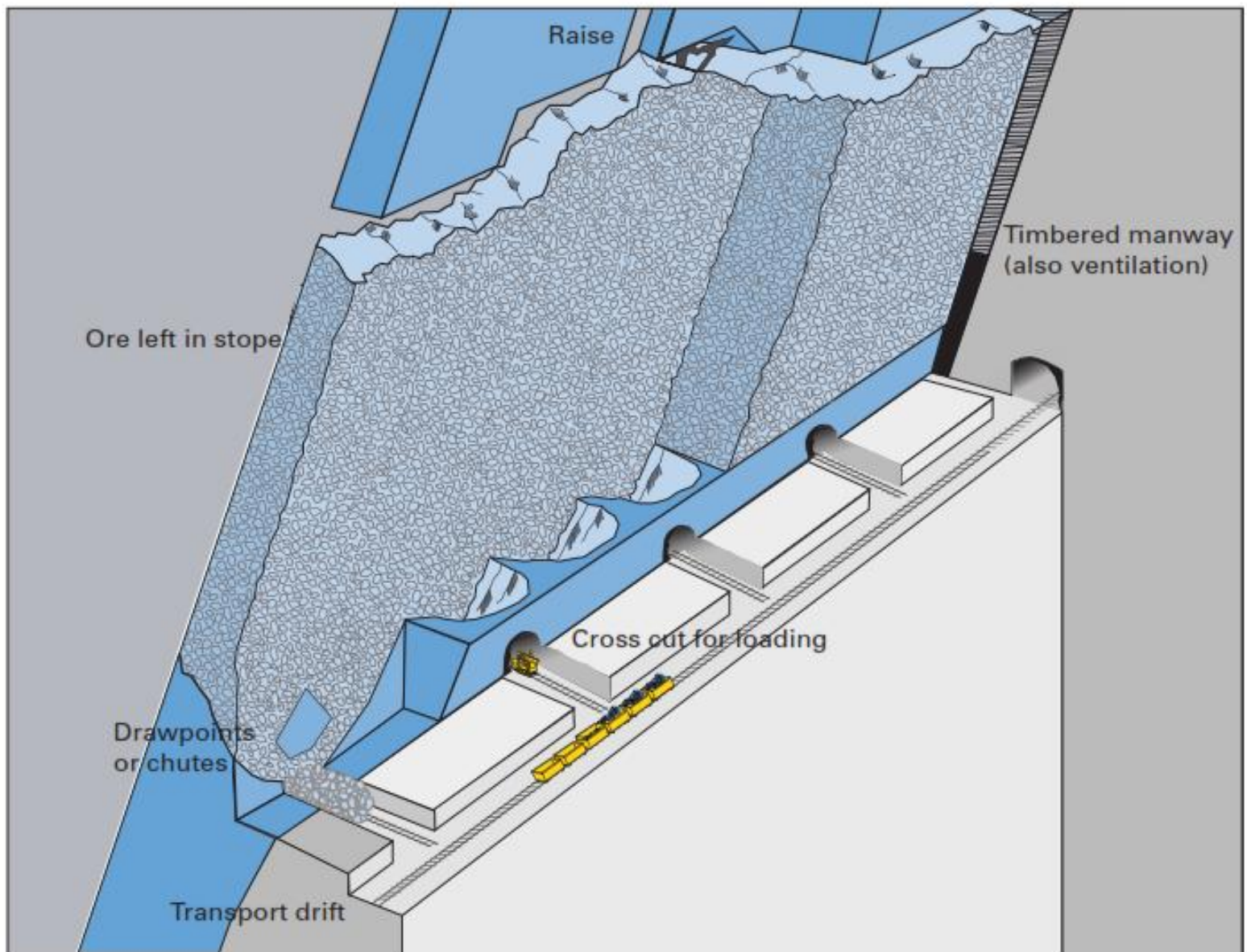
Transport infrastructure: Shaft, Cage, Skip (for ore transportation), Utility vehicle, LHD, Decline ramp, Truck, Conveyor

Services: Electric power for Power drill rigs, pumps and other electric operated machines, water reticulation, Ventilation, air quality

Raises and Winzes: raise boring, The raise boring machine (RBM) may be used for boring ventilation raises, ore-passes, rock fill passes, and slot raises.

Underground machineries: raise boring machine (RBM), Mechanized bolting, Mechanized scaling

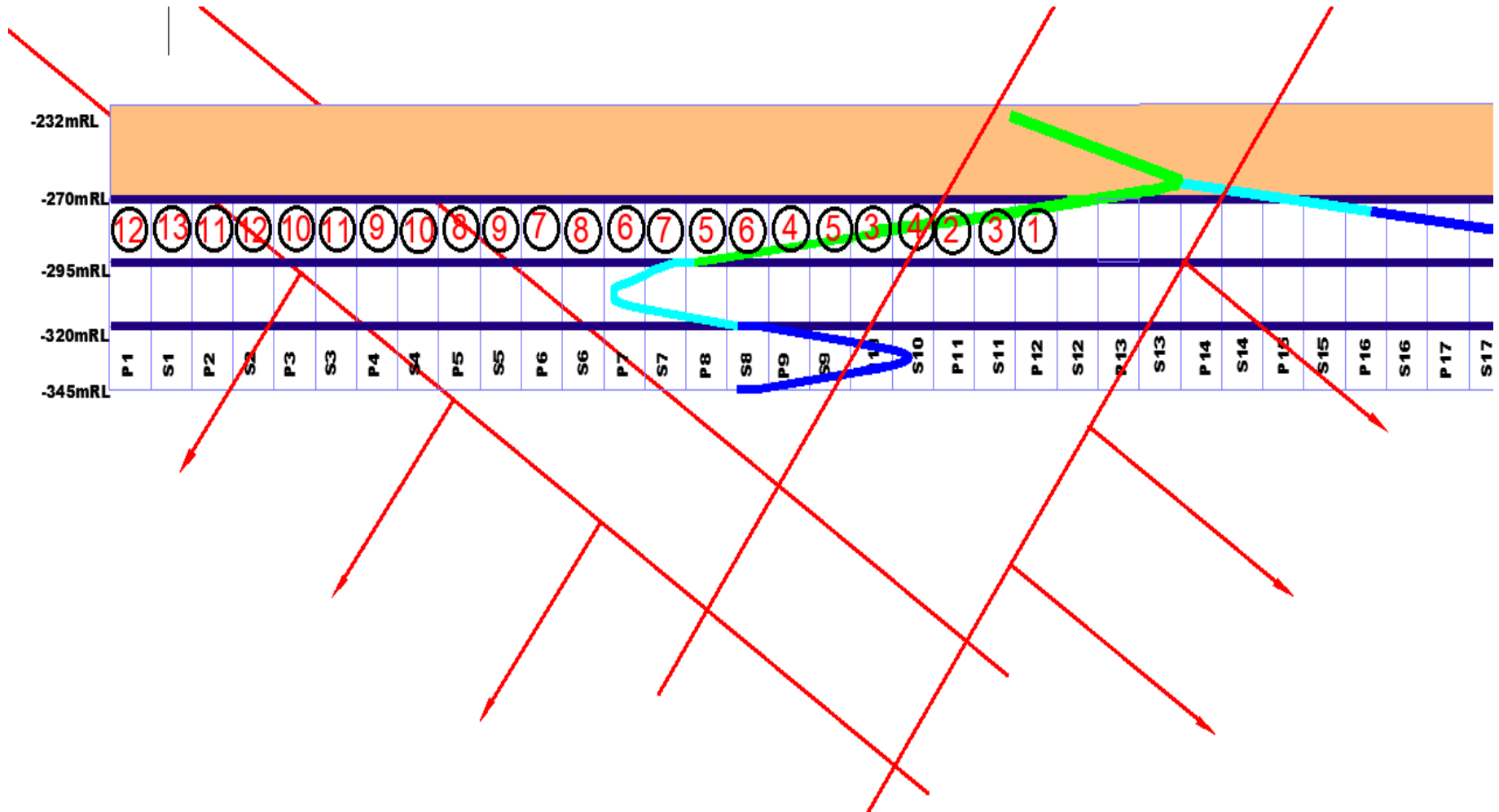




Shrinkage stoping layout.

Underhand mining

Proposed Mining sequence to reduce the stress damage in FWD and enable crown pillar recovery.



Major equipment's, SKM

- Low Profile Dump Truck 50t capacity : 14 Nos
- Low Profile Dump Truck 30t capacity : 4 Nos
- Load Haul Dump 17t capacity : 8 Nos
- Load Haul Dump 10t capacity : 8 Nos
- Single/Double Boom Drill Jumbos : 13 Nos
- Electro Hydraulic Production Drill : 3 Nos
- Specialized Equipment
 - Motor Grader : 2 Nos
 - Rock Bolter : 1 No
 - Mobile Carrier Exploration Rig : 1 No
 - V30 Slot Raising Drill Rig : 1 No
 - Charmec : 1 No
 - Scaler : 1 No
- Service Vehicle:
 - Personnel Carrier : 7 Nos
 - Lifting Equipment (Scis./ Boom Lift/ Pipe Handler) : 8 Nos
 - RBO & Rexton : 8 Nos
 - Explosive Carriers : 2 Nos
 - Diesel Bowser : 1 No
 - Water Sprinkler : 1 No

Mining Operations - Best Practices 1

Trackless Loading Hauling



Mechanized Mine Development



Bulk Emulsion Charging



Mobile Carrier Exploration Rig



Mining Operations - Best Practices 2

Refuge Chamber



Simulator



V30 Slot Raising Machine

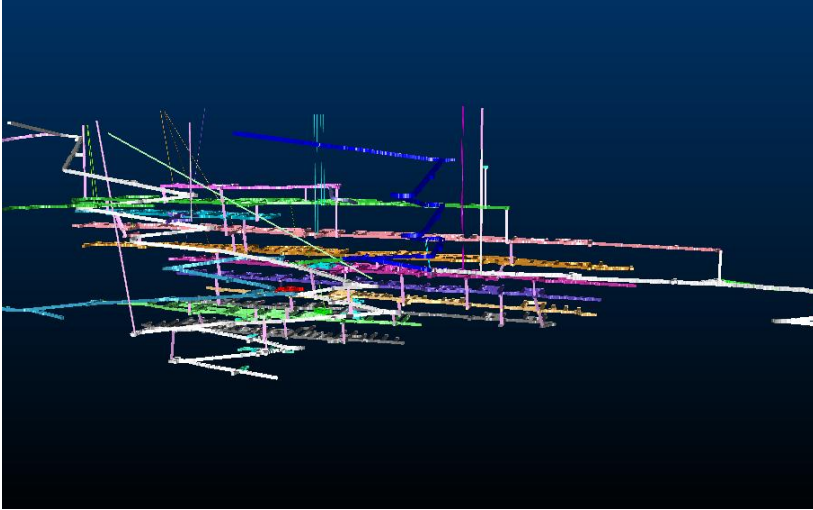


Scaler



Mining Operations - Best Practices 4

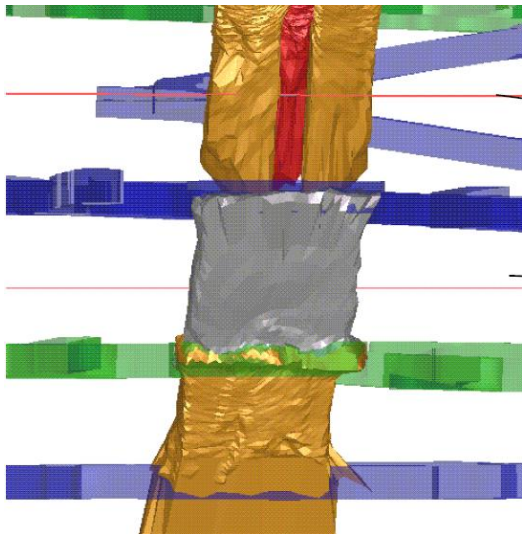
Latest Software



Simulator



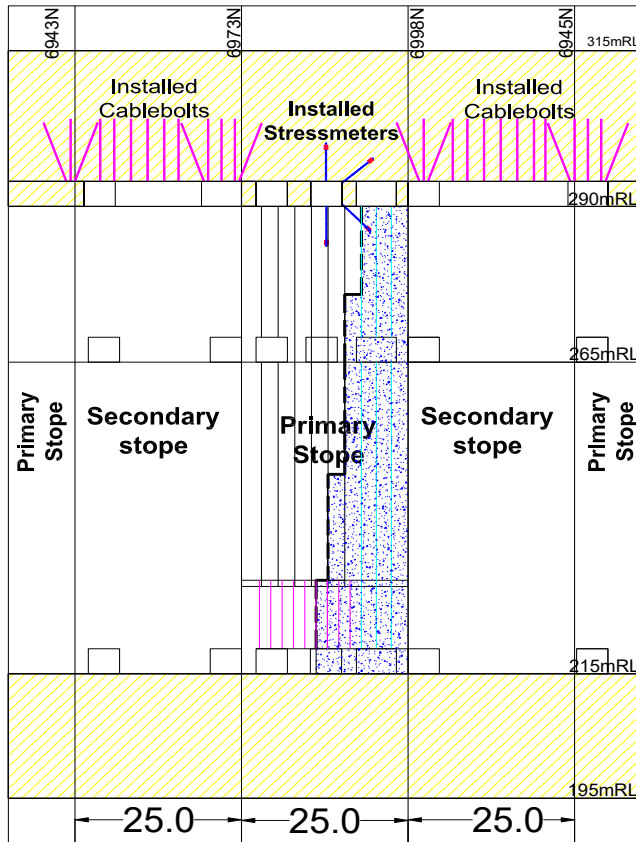
Cavity Monitoring System



Sirovision



Sindesar Khurd Mine – Mining Method, BHS with paste



LVS

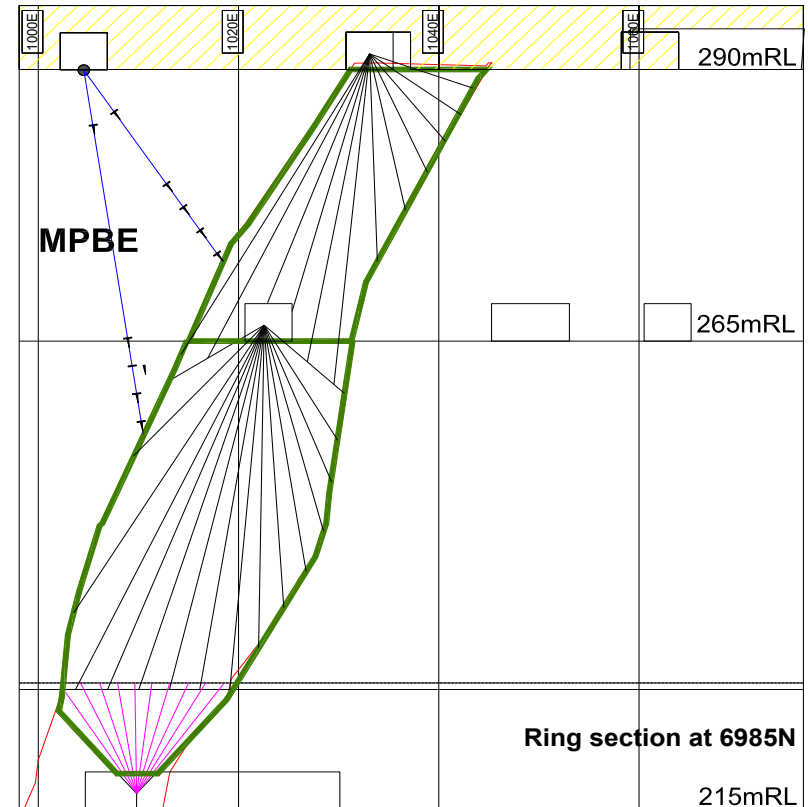
Stope Configuration:-

Strike length: 25m to 35m

Average width: 15m to 55m

Height: 75m

Tonnage: 200000t to 500000t



Drill Hole Diameter:-

Slot: 165mm

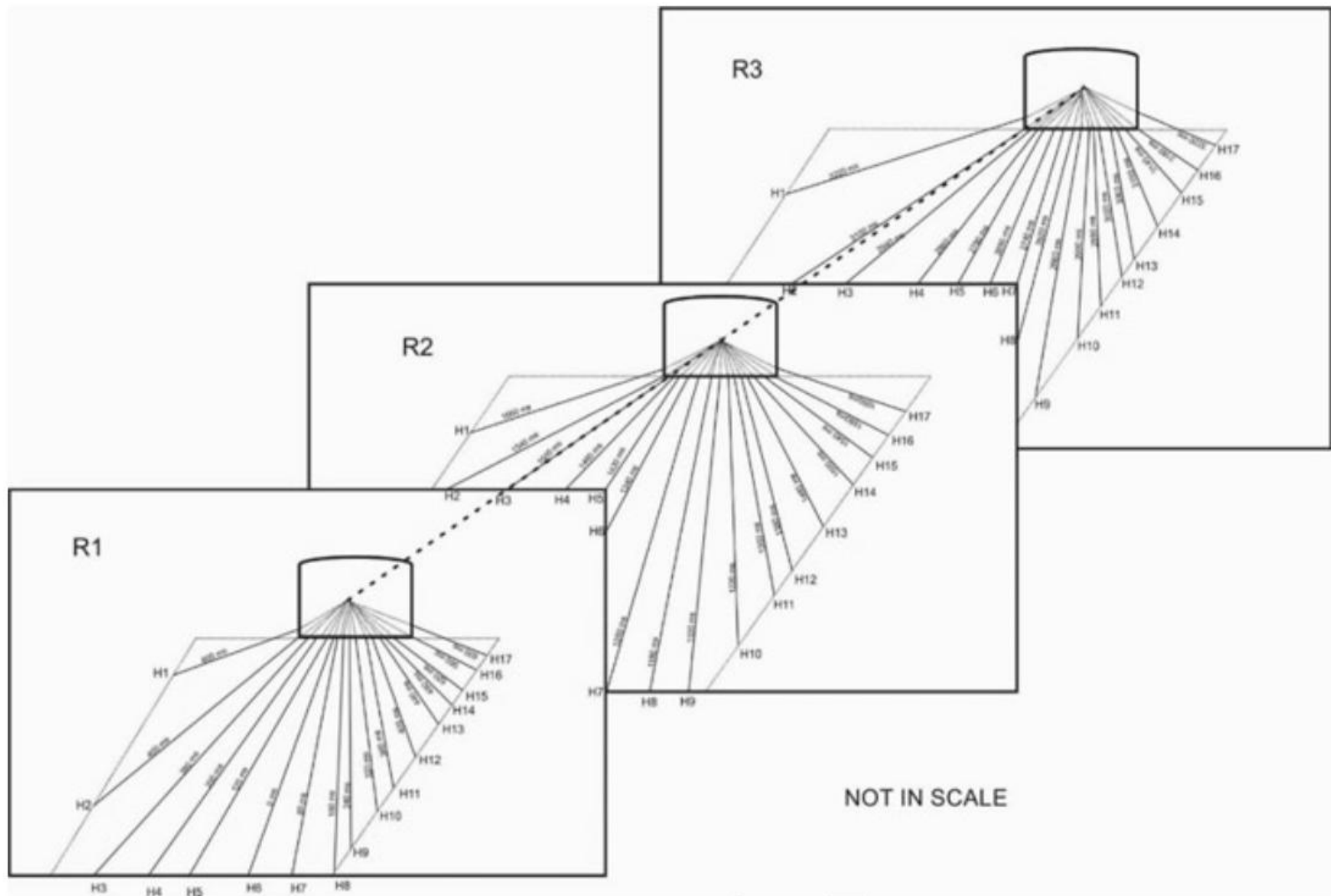
Ring: 115/102mm

Trough: 65mm

Ring Burden:- 2.65m

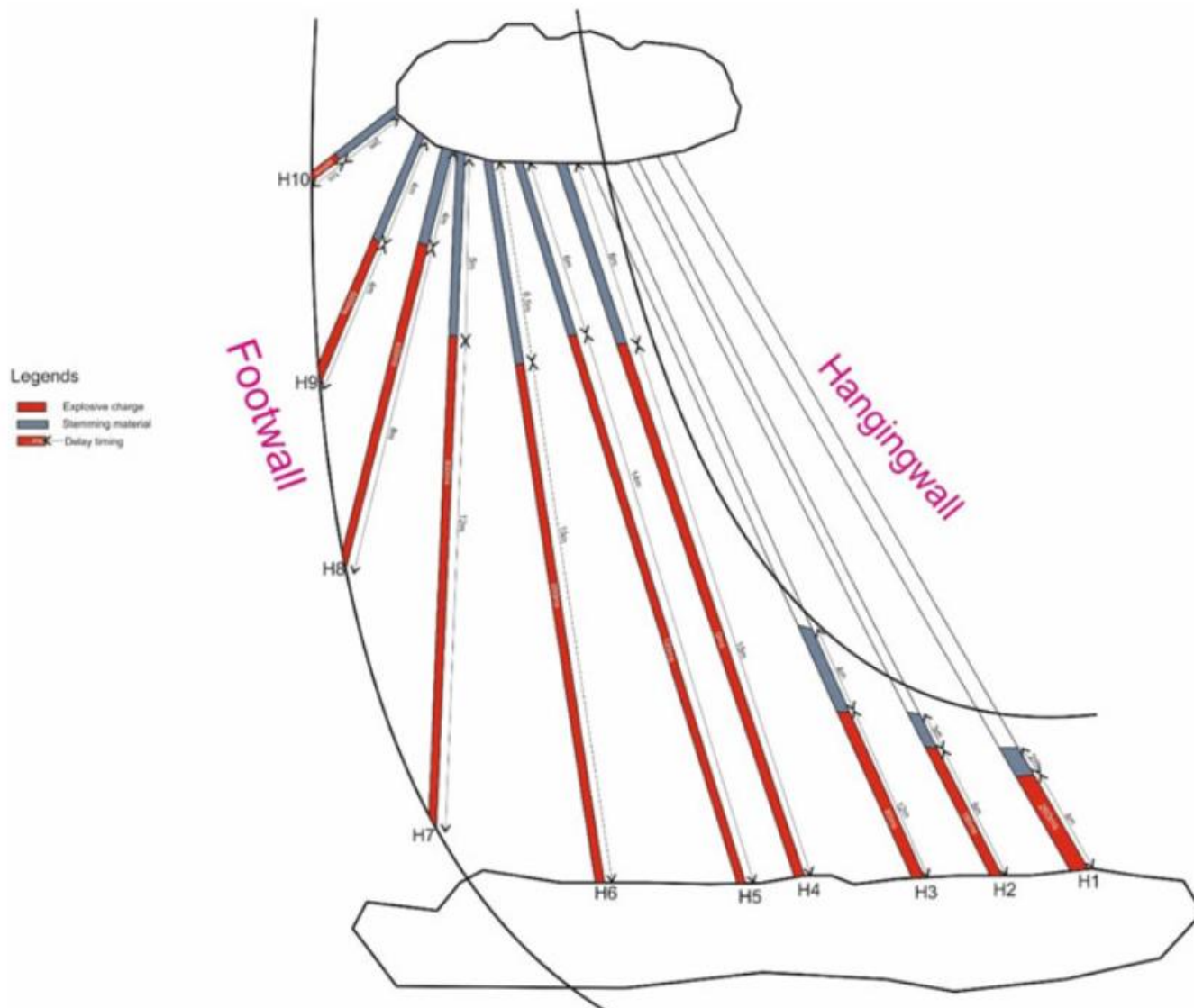
Toe Spacing:- 3.0m

Multi ring blasting



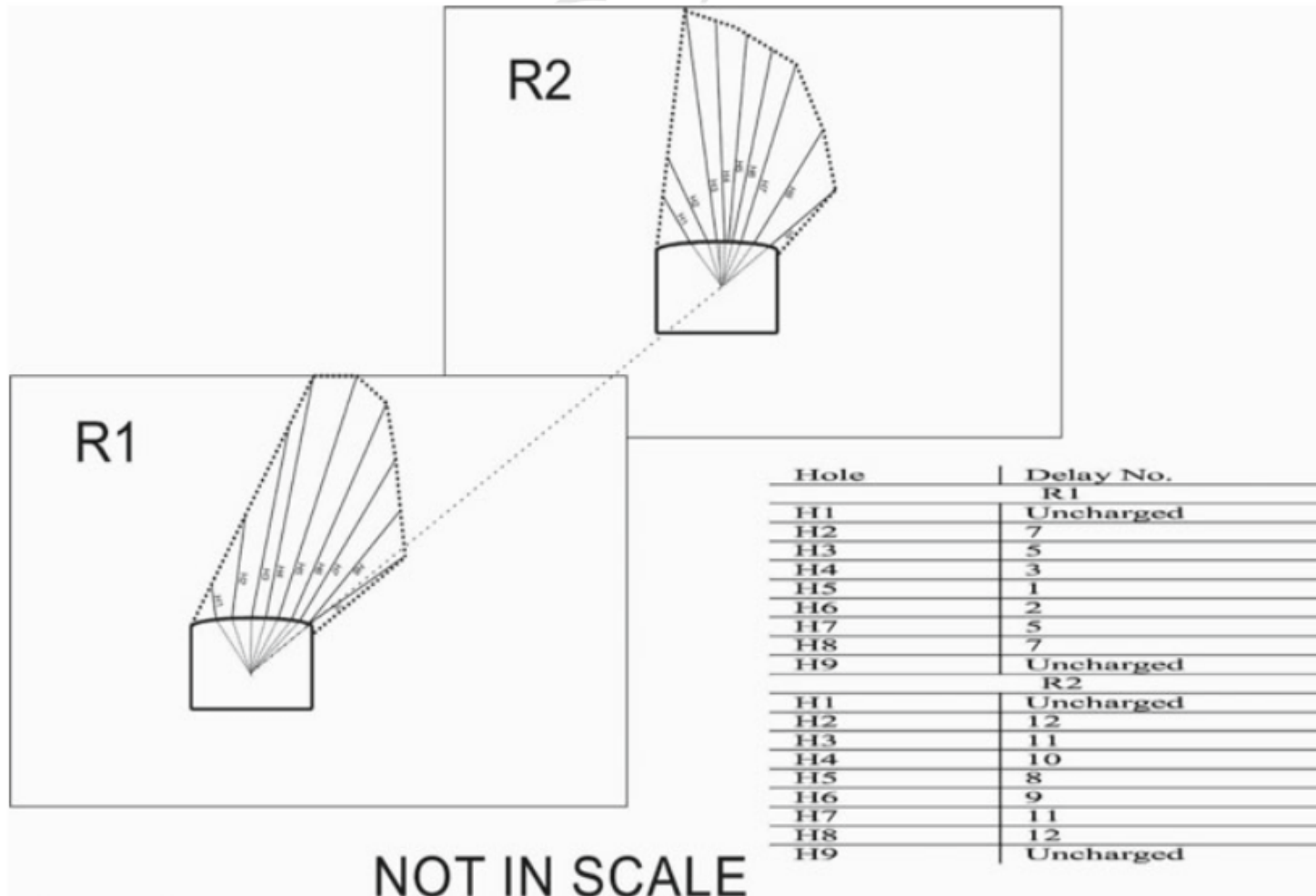
Himanshu et al. (2024)

Ring blasting

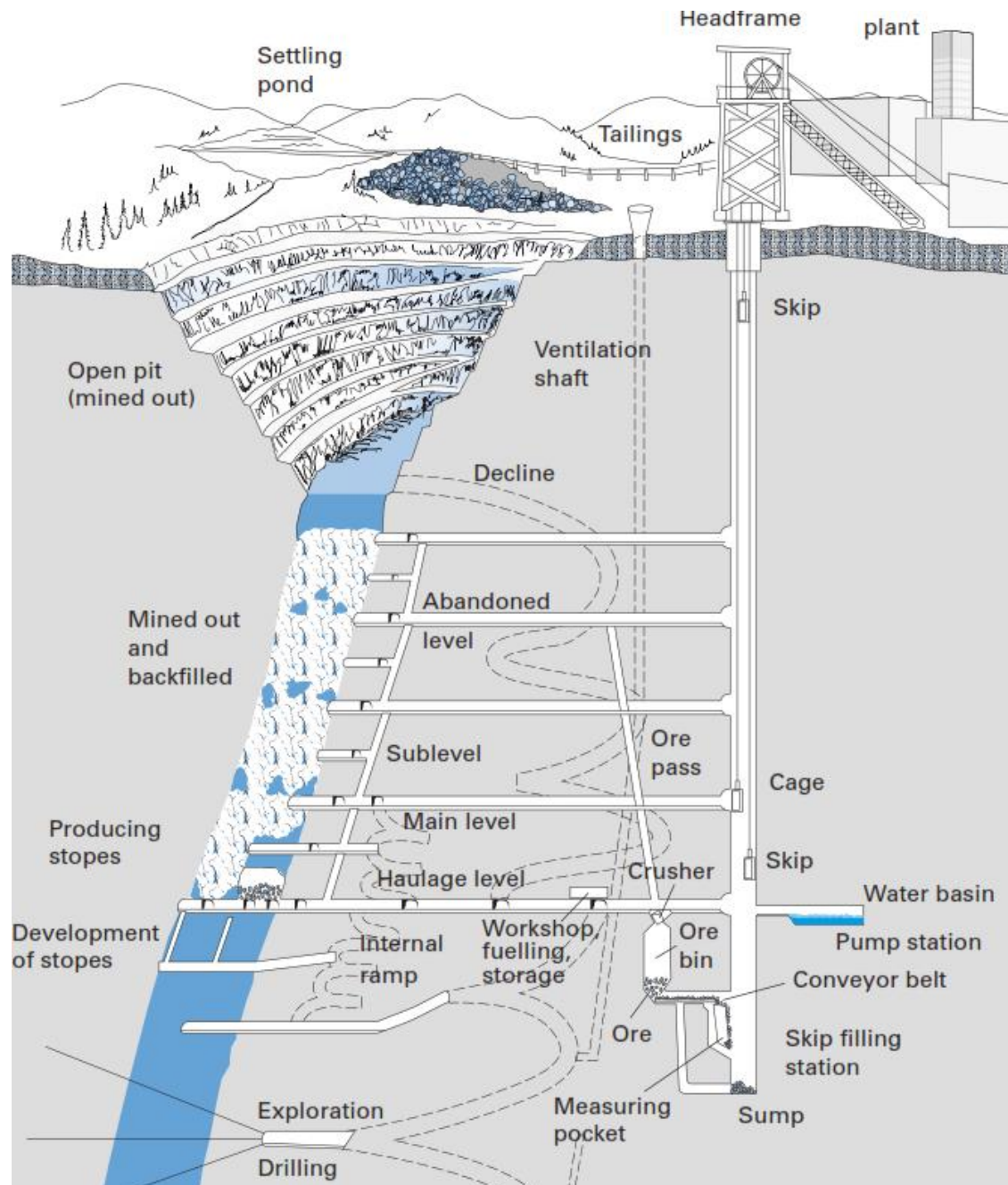


Himanshu et al. (2024)

Up hole ring blasting



Himanshu et al. (2024)



3D model of a typical underground mine below opencast mine

